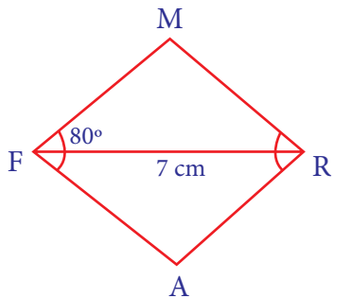
5.14.4 Construction of a rhombus when one diagonal and one angle are given

Example 5.37

Construct a rhombus FARM with FR = 7 cm and $\angle F = 80°$. Also find its area.



Rough diagram

Solution:

Given: FR = 7 cm and $\angle F = 80°$

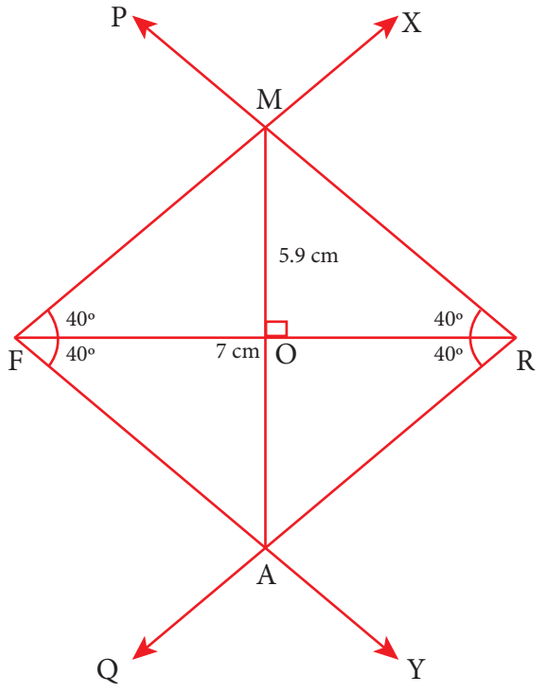


Fig. 5.56

Steps:

1. Draw a line segment FR = 7 cm.
2. At F, make $\angle RFX = \angle RFY = 40°$ on either side of FR.
3. At R, make $\angle FRP = \angle FRQ = 40°$ on either side of FR.
4. Let FX and RP cut at M and FY and RQ cut at A.
5. FARM is the required rhombus.

Calculation of Area:

Area of rhombus FARM $= \frac{1}{2} \times d_1 \times d_2 \; sq.units$

$= \frac{1}{2} \times 7 \times 5.9 = 20.65 \; sq.cm$

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